



Biogenic uraninite precipitation and its reoxidation by iron(III) (hydr)oxides: A reaction modeling approach

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Received 21 September 2010; accepted in revised form 25 April 2011; available online 19 May 2011

Abstract

One option for immobilizing uranium present in subsurface contaminated groundwater is *in situ* bioremediation, whereby dissimilatory metal-reducing bacteria and/or sulfate-reducing bacteria are stimulated to catalyze the reduction of soluble U(VI) and precipitate it as uraninite (UO₂). This is typically accomplished by amending groundwater with an organic electron donor. It has been shown, however, that once the electron donor is entirely consumed, Fe(III) (hydr)oxides can reoxidize biogenically produced UO₂, thus potentially impeding cleanup efforts. On the basis of published experiments showing that such reoxidation takes place even under highly reducing conditions (e.g., sulfate-reducing conditions), thermodynamic and kinetic constraints affecting this reoxidation are examined using multicomponent biogeochemical simulations, with particular focus on the role of sulfide and Fe(II) in solution. The solubility of UO₂ and Fe(III) (hydr)oxides are presented, and the effect of nanoscale particle size on stability is discussed. Thermodynamically, sulfide is preferentially oxidized by Fe(III) (hydr)oxides, compared to biogenic UO₂, and for this reason the relative rates of sulfide and UO₂ oxidation play a key role on whether or not UO₂ reoxidizes. The amount of Fe(II) in solution is another important factor, with the precipitation of Fe(II) minerals lowering the Fe⁺² activity in solution and increasing the potential for both sulfide and UO₂ reoxidation. The greater (and unintuitive) UO₂ reoxidation by hematite compared to ferrihydrite previously reported in some experiments can be explained by the exhaustion of this mineral from reaction with sulfide. Simulations also confirm previous studies suggesting that carbonate produced by the degradation of organic electron donors used for bioreduction may significantly increase the potential for UO₂ reoxidation through formation of uranyl carbonate aqueous complexes.

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1. INTRODUCTION

Bioreduction of soluble U(VI) to U(IV) as solid-phase UO₂ using dissimilatory metal-reducing bacteria including sulfate-reducing bacteria (SRB) (Lovley and Phillips, 1992; Beyenal et al., 2004; Sani et al., 2004, 2005, 2006) is currently

being considered as a remediation alternative at several uranium contaminated sites (e.g., Anderson et al., 2003; Istok et al., 2004, 2010; Vrionis et al., 2005; Wu et al., 2006). It has been shown, however, that UO₂ can readily reoxidize in the presence of various oxidizing agents, including oxygen (Abdelouas et al., 1999; Zheng et al., 2002; Moon et al., 2007; Senko et al., 2007; Komlos et al., 2008), nitrate and other nitrogen species (Finneran et al., 2002; Istok et al., 2004; Moon et al., 2007), Mn oxides (Fredrickson et al., 2002; Liu et al., 2002), and most notably Fe(III) (hydr)oxides (Nevin and Lovley, 2000; Wan et al., 2005; Sani et al., 2004, 2005; Senko et al., 2007; Ginder-Vogel et al.,

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2006, 2010). Reoxidation of UO₂ by Fe(III) (hydr)oxides has received particular attention due to the ubiquitous presence of these minerals in the subsurface, and because the reoxidation of UO₂ in this case has been shown to occur even under sulfate-reducing conditions (Sani et al., 2004, 2005).

The thermodynamic viability of UO₂ oxidation by Fe(III) (hydr)oxides has been assessed previously (Ginder-Vogel et al., 2006), showing favorability primarily under mildly reducing conditions in the presence of ferrihydrite, and also depending on solution composition, particularly on Fe(II) and inorganic carbon concentrations. The potential for Fe(II) to reduce U(VI) has long been recognized, particularly when catalyzed by sorption of Fe(II) onto Fe(III) (hydr)oxides (Liger et al., 1999; Behrends and Van Cappellen, 2005; Jeon et al. 2005; Jang et al., 2008; Regenspurg et al., 2009; Chakraborty et al., 2010). The role of aqueous Fe(II) and Fe(II) minerals as potential UO₂ anti-oxidants has also been evaluated (Abdelouas et al., 1999; Zhong et al., 2005), although only for systems where O₂ is the main oxidant.

These studies have shown that the reduction and possible reoxidation of UO₂ in the subsurface proceed following complex biotic and abiotic pathways, with outcomes that may not be intuitive or fully understood. For example, it would be anticipated from the decreasing thermodynamic stability of hematite, goethite, and ferrihydrite (Navrotsky et al., 2008) that the reoxidation of biogenic UO₂ by these minerals would yield respectively higher U(VI) concentrations in solution. Such behavior was indeed reported in batch experiments carried out by Ginder-Vogel et al. (2006) using biogenic UO₂ prepared separately from reoxidation experiments. However, Sani et al. (2004, 2005) reported the reverse behavior when reoxidation took place following bioreduction under electron-donor-limited sulfate-reducing conditions (upon depletion of the electron donor, in one same experiment, i.e., same incubation). In this case, U(VI) concentrations were higher in solutions containing hematite than in those containing ferrihydrite and goethite, suggesting increasing U(VI) concentrations with more thermodynamically stable phases. One potential explanation for this behavior suggested by Sani et al. was reductive dissolution of ferrihydrite by biogenic sulfide (produced during the UO₂ bioreduction period). The reductive dissolution by sulfide would proceed to a greater extent with ferrihydrite compared to systems containing hematite or goethite, thus potentially depleting the Fe(III) pool available for UO₂ reoxidation. The progressively higher surface areas of hematite, goethite, and ferrihydrite could also lead to differences in U(VI) adsorption, and thus differences in UO₂ reoxidation behavior (e.g., Ginder-Vogel et al., 2010).

Several other complicating factors affect the evaluation of reoxidation behavior. The formation of uranyl carbonate complexes due to production of carbonate resulting from microbial degradation of the organic electron donor likely promotes reoxidation (Tokunaga et al., 2008). Furthermore, thermodynamic analyses are complicated because the small particle size (1–5 nm) of biogenic UO₂ (Suzuki et al., 2002; Sani et al., 2006; Bargar et al., 2008; Sharp

et al., 2009) is expected to affect solubility (Banfield and Zhang, 2001; Opel et al., 2007). Also, the thermodynamic stability of Fe(III) (hydr)oxides (i.e., derived from caloric measurements) may not reflect actual solubilities because of surface hydration (Jang et al. 2007a, 2008). Adding to these complexities is the formation of secondary Fe(II) minerals such as siderite, magnetite, and iron sulfides, as these affect Fe(II) activity in solution, potentially also impacting UO₂ reoxidation by Fe(III) (hydr)oxides.

Building on previous studies, particularly those of Sani et al. (2004, 2005), the goal of this study is to further investigate potentially key thermodynamic and kinetic factors affecting the reoxidation of biogenic UO₂ by Fe(III) (hydr)oxides, and to develop a reaction network to model U bioreduction with concomitant reoxidation by Fe(III). This is accomplished using rather simplified multicomponent (bio)geochemical computations that are intended to elucidate operative reactions in real systems without necessarily capturing the full complexity of these systems. The thermodynamic stability of various phases involved is reviewed. The effect of particle size on solubility, and the limits of biogenic UO₂ solubility as a function of pH are also estimated for systems containing various Fe(III) (hydr)oxides. The focus is then shifted to sulfide oxidation by Fe(III) (hydr)oxides (i.e., reductive dissolution by sulfide), which is thermodynamically more favorable than for biogenic UO₂, and thus competes with biogenic UO₂ oxidation. Finally, a simple biogeochemical reaction network is developed for application to the experiments of UO₂ bioreduction and reoxidation reported by Sani et al. (2004).

It should be noted that the actual mechanisms of UO₂ oxidation by Fe(III) (hydr)oxides have not been fully identified. Thermodynamically, the concentrations of free U(IV) (e.g., Langmuir, 1978) and free Fe(III) (e.g., Millero et al., 1995) at equilibrium with UO₂ and Fe(III) hydroxides at circumneutral pH are extremely small, suggesting the oxidation takes place on mineral surfaces rather than in solution. The reoxidation could occur by the release and migration of Fe(III) from Fe(III) (hydr)oxides to the UO₂ surface, with reduction of Fe(III) on the UO₂ surface, or conversely by the release and migration of U(IV) from UO₂ to the Fe(III) solid surface, thus with oxidation of U(IV) on the Fe(III) solid. Ginder-Vogel et al. (2010) suggest that the latter is most likely to occur, with an overall rate of reoxidation that is controlled by the rate of UO₂ dissolution. The thermodynamic and kinetic modeling in this study does not distinguish between these various mechanisms because it is based solely on reaction affinity, which is path independent in a closed system. Therefore, it cannot (nor is it intended to) shed light on actual reaction mechanisms per se. Nevertheless, despite being non-mechanistic in nature, the model allows the assessment of complicated thermodynamic balances and kinetic effects that ultimately define whether a reaction mechanism can operate or not.

2. MODELING CODE AND THERMODYNAMIC DATA

Simulations presented in this paper were carried out using TOUGHREACT (Xu et al., 2011). Other codes were also used for

preliminary and/or confirmatory analyses (PHREEQC, Parkhurst and Appelo, 1999; and CHILLER, Reed, 1982, 1998). Thermodynamic data were taken from a database (SNL, 2007) derived in large part from Shock et al. (1997) and Helgeson et al. (1978), which was then updated as necessary for this study (see relevant subset in Appendix A). Data for uranium species and minerals were taken from SNL (2007) if within error margins reported in the compilation of Guillaumont et al. (2003), or replaced with the data from Guillaumont et al. (2003) if inconsistent with this data set (Appendix A). These data were augmented with those from Dong and Brooks (2006) for U(VI)–Ca–CO₃ complexes. Other data were included in the database as shown in Appendix A, notably for Fe–S aqueous species and solids (Rickard, 2006). The type and solubility product of UO₂ and Fe(III) (hydr)oxides is key to both modeling results and the thermodynamic viability of UO₂ reoxidation. For this reason, available thermodynamic data for these phases were given particular attention, as discussed further below.

Schoepite (UO₃·2H₂O) was predicted to precipitate in some of the simulations. Because the equilibrium constant given by Guillaumont et al. (2003) is on the low range of reported values for this mineral, simulations in which schoepite was predicted to form were also run using meta-schoepite instead, taking thermodynamic data for this mineral from Gorman-Lewis et al. (2008). These data yield a solubility for meta-schoepite that is almost one order of magnitude larger than the solubility of schoepite.

2.1. UO₂ solubility

The solubility of crystalline (UO_{2(c)}) and amorphous uraninite (UO_{2(am)}) has been a matter of debate because of the formation of amorphous coatings on crystalline UO₂ and the difficult control of redox potential in experiments (e.g., Rai et al., 2003). Solubilities critically reviewed and selected by Guillaumont et al. (2003) for both UO_{2(c)} and UO_{2(am)} are generally well accepted and are adopted in this study ($\log(K_{sp}) = -60.86 \pm 0.36$ and -54.5 ± 1 , respectively, for the reaction: $\text{UO}_2 + 2\text{H}_2\text{O} \leftrightarrow \text{U}^{+4} + 4\text{OH}^-$). In a more recent study, Opel et al. (2007) measured the particle size-dependent solubility of crystalline and amorphous UO₂. These authors avoided the potential problems of previous studies by electrochemically reducing U(VI) solutions to U(IV), and report solubilities ($\log(K_{sp}) = -59.6 \pm 1$ and -54.1 ± 1 for UO_{2(c)} and UO_{2(am)}, respectively) consistent with those selected by Guillaumont et al. (2003). Uncertainties remain, however, about the solubility of biogenic uraninite (UO_{2(bio)}). Previous studies have shown UO_{2(bio)} to consist of nanoparticulate (typically 1–5 nm) stoichiometric UO₂ (e.g., Suzuki et al., 2002; Schofield et al., 2008; Sharp et al., 2009). The large surface area effects at such small particle sizes could increase solubility considerably (Suzuki et al., 2002). However, recent solubility measurements of both synthetic (bulk) and nanoparticulate UO_{2(bio)} (Ulrich et al., 2008) suggest that the surface area effect may be smaller than anticipated, with solubility generally (but not always) within the magnitude of reported UO_{2(am)} solubility values. In other experiments (Senko et al., 2007), the formation of UO_{2(bio)} at rapid rates of U(VI) reduction produced particles that were smaller and less aggregated, and that subsequently underwent more reoxidation, than UO_{2(bio)} particles formed at slower rates, evidencing the strong control of size and aggregation state on reoxidation. In light of these various findings, simple calculations were carried out (Appendix B) to evaluate the effect of particle size on the solubility of UO₂ and to put these data in perspective with generally accepted solubilities for UO_{2(c)} and UO_{2(am)} (Fig. B1). These show that at a particle size of ~3 nm, the estimated solubility product of UO₂ is consistent with the solubility value reported for UO_{2(am)} by Guillaumont et al. (2003), and also show that at such small particle size, a 1 nm change in particle diameter yields a change of about two orders of

magnitude in solubility (Fig. B1). The specific method applied to evaluate solubility as a function of particle size, including assumptions of particle shape, as well as inclusion (or not) of surface hydration and/or stress effects also yield significant differences in results. Given these uncertainties, the solubility adopted here for UO_{2(bio)} is taken as that estimated for 3 nm particles ($\log(K_{sp}) = -54.6$), in line with the accepted solubility data for UO_{2(am)} and with values used by previous investigators (e.g., Ulrich et al., 2008; Ginder-Vogel et al., 2006). It should be noted, however, that this solubility could actually be much greater (or not) depending on actual particle size and aggregation.

2.2. Solubility of Fe(III) (hydr)oxides

The solubility of hematite (Fe₂O₃), goethite (FeOOH), and ferrihydrite (6-line, expressed as Fe(OH)₃) adopted in this study (Appendix A) are consistent with Gibbs free energy values reported in Majzlan et al. (2004) for these minerals, and in Diakonov et al. (1999) for Fe(OH)₄[−]. These data are also within error margins of solubilities reported by Diakonov et al. (1999). The Fe(III) hydrolysis constants (Shock et al., 1997) in the adopted database are also consistent with those re-evaluated by Stefansson (2007) (essentially within the error margins of these data). It has been shown that the solubility of more stable Fe(III) (hydr)oxides, such as hematite, increases to values in the range of ferrihydrite solubility as particle size decreases (Navrotsky et al., 2008). The difference in solubility product between hematite and ferrihydrite, as implemented in this study, would correspond to a decrease in particle size from bulk hematite to around 4 nm when applying Eq. (B.1) with the surface energy for hematite reported by Navrotsky et al. (2008) (0.75 J/m² for hydrated surfaces, and neglecting surface stress effects). Therefore, by considering Fe(III) (hydr)oxide solubilities spanning a range from hematite to ferrihydrite solubilities, the increase in solubility with decreasing particle size can be regarded as being indirectly considered. It should also be noted that the thermodynamic solubility of hematite may not be representative of actual solubility (Jang et al., 2007a). These authors report that because of hydration, the Fe(III) content and pH of solutions equilibrated with hematite at low pH can evolve (within one to a few months) towards values reflecting equilibration with ferrihydrite, even though solids are invariably identified as hematite. Note that the stability of magnetite is also considered because, as discussed later, it can form by microbial reduction of Fe(III) (hydr)oxides. It is treated here as Fe₃O₄ (equivalent to FeO:Fe₂O₃), using free energy data from Helgeson et al. (1978).

3. SPECIATION AND EQUILIBRIUM SOLUBILITY IN THE U–Fe–Ca–CO₃ SYSTEM

To assess thermodynamic limits of UO_{2(bio)} reoxidation, as well as U and Fe speciation, equilibrium computations were run by reacting UO_{2(bio)} with (each separately) hematite, goethite, magnetite, and ferrihydrite (Fig. 1). These calculations were performed for solutions containing initial concentrations of 0.1 mM Ca and 3 mM total dissolved carbonate (in the range typically encountered in groundwater) and over a pH range of 5–9. The solutions were first equilibrated with UO_{2(bio)} at 25 °C and pH 9 under fully reducing conditions ($\log(f\text{O}_2) < -80$), then allowed to equilibrate with the specified Fe(III) mineral (using NaOH to set the initially alkaline pH). The pH of the solution was then lowered by numerically titrating acid (as HCl) and maintaining equilibrium of the solution with both UO_{2(bio)} and the selected Fe(III) (hydr)oxide. Calcite, siderite, and

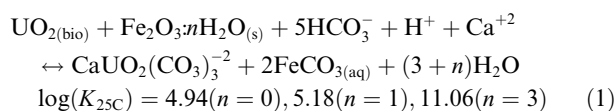
schoepite were allowed to form in all cases, but were not assumed initially present. The solution was maintained at equilibrium with UO_{2(bio)} and each specified Fe(III) mineral over the entire simulated pH range (i.e., unlimited supply of these minerals). Note that the total dissolved concentrations of calcium and carbonate were allowed to decrease upon precipitation of calcite and/or siderite.

Results show a wide range of total dissolved U concentrations attributed essentially entirely to U(VI) (Fig. 1). Dominance of U(V) (as UO₂⁺) is computed over a very narrow pH range near 5.8 when ferrihydrite is the oxidant (Fig. 1C), which is consistent with earlier U speciation modeling work presented by Langmuir (1978). Experimental data for this system under identical abiotic conditions were not available for direct comparison with model results. However, Ginder-Vogel et al. (2006) measured U and Fe concentrations in similar systems without Ca (i.e., at pH 7, 3 mM dissolved carbonate with UO_{2(bio)} and either hematite, goethite or ferrihydrite). Modeled and measured data points for these conditions are also plotted on Fig. 1. These show reasonable agreement (within a factor <3) for the hematite and goethite systems. For the ferrihydrite system, however, measured U concentrations are about 100× smaller than modeled values, with Fe concentrations smaller by a factor ~4 (Fig. 1C). This suggests reoxidation far from equilibrium conditions, and/or possible evolution towards a system where ferrihydrite transforms into magnetite, as further discussed below.

In all cases (Fig. 1), calcite forms at pH >~8. Siderite forms only when ferrihydrite is reacted, and over the entire investigated pH range (Fig. 1C). Schoepite (as UO₃·2H₂O) is predicted to form below pH ~5.5–6 when either hematite, goethite, or magnetite are reacted, and below pH ~7.4 when ferrihydrite is the reactant. When schoepite is not allowed to form, meta-schoepite solubility is reached only when ferrihydrite is reacted at pH below ~6.4, or when magnetite is reacted at pH below ~5.2. The precipitation of either schoepite or meta-schoepite results in a discontinuity in the computed U solubility curve. Note that equilibrium with UO_{2(bio)} is maintained over the entire simulated pH range for all cases.

Expressing hematite, goethite, and ferrihydrite as Fe₂O₃, Fe₂O₃·H₂O (i.e., 2FeOOH), and Fe₂O₃·3H₂O (i.e., 2Fe(OH)₃), respectively, the overall oxidation reactions taking place in this system can be written as follows:

Alkaline region:



Circum-neutral region:

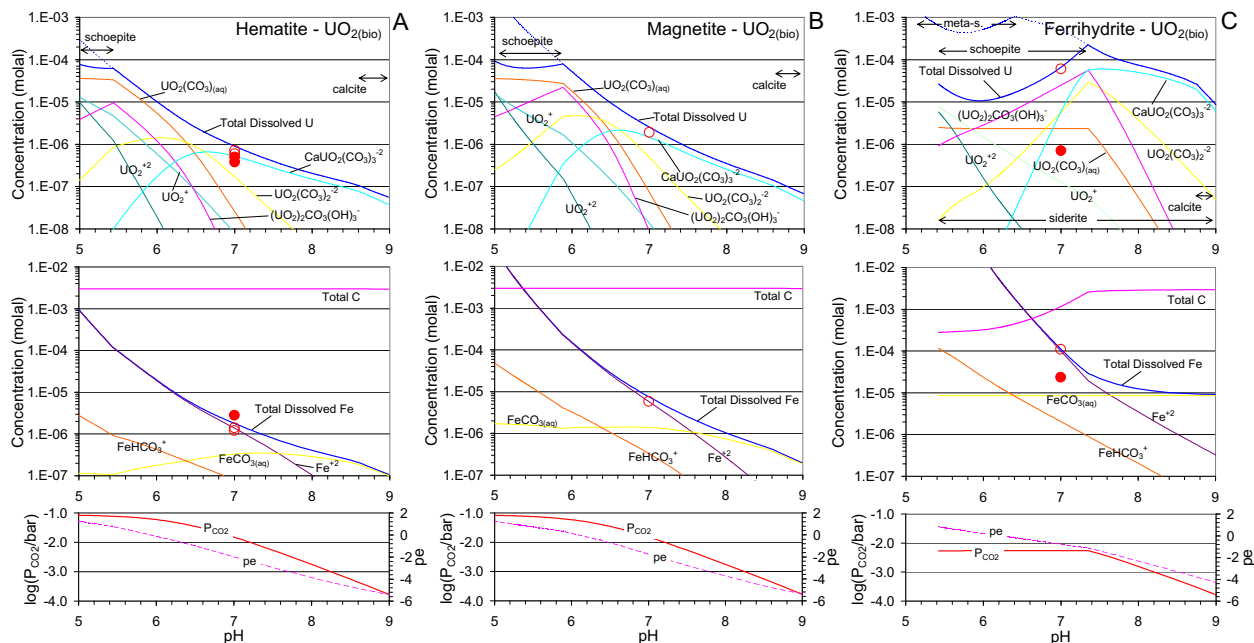
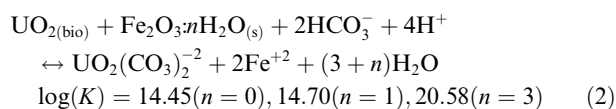
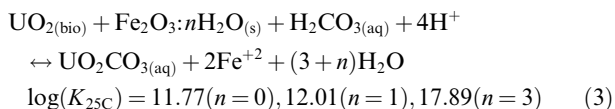
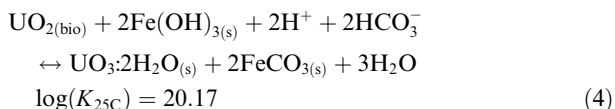


Fig. 1. Computed total dissolved U and Fe concentrations upon reaction (at equilibrium) of UO_{2(bio)} with (separately) hematite (A), magnetite (B), and ferrihydrite (C) (in initially fully reduced solutions with initial 0.1 mM Ca and 3 mM ΣCO₃ concentrations). Most dominant individual U and Fe species are also shown. Reaction with goethite yields solubilities similar to, and about 20% higher, than in the hematite case (data not shown). Total dissolved U concentrations calculated using meta-schoepite instead of schoepite are shown with dotted lines. Symbols indicate measured (solid circles; Ginder-Vogel et al., 2006) and modeled (open circles) data for a similar system without Ca (with data for both the hematite and goethite systems plotted in (A), the higher values representing the goethite system; note that measured Fe in (A) was below detection limits for the hematite case and is shown only for the goethite case). Double arrows show the pH range in which the specified mineral is computed to form.

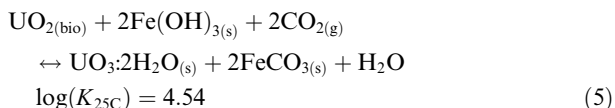
Acid region:



Note that when ferrihydrite is the reactant, the formation of both siderite and schoepite below pH ~ 7.5 yields an oxidation reaction that is solely dependent on pH and bicarbonate concentration:



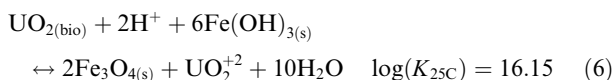
At low pH, as HCO_3^- fully protonates, reaction (4) becomes solely dependent on total dissolved carbonate concentration (univariant). The equilibrium p_{CO_2} imposed by this reaction is fixed (Fig. 1C):



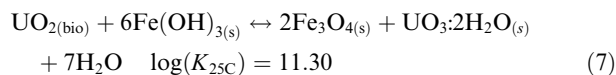
at a value $\sim 10^{-2.3}$ ($\log(f_{\text{CO}_2}) = -\log(K)/2 \sim \log(p_{\text{CO}_2})$). Therefore, any externally applied CO_2 partial pressure higher than this value readily drives reoxidation to full consumption of $\text{UO}_{2(\text{bio})}$. CO_2 partial pressures above this value are quite common in soils with typical biological activity levels (e.g., Coudrain-Ribstein et al., 1998), and could get much higher upon stimulated bioreduction (i.e., $\log(p_{\text{CO}_2})$ around -1 in the bioreduction simulations presented later, for a closed system).

Reactions similar to reactions (1)–(3) can also be written when magnetite is the reactant (Fig. 1B). Even though magnetite contains Fe(II), its higher solubility than either hematite or goethite in the present system yields higher dissolved U concentrations than when hematite or goethite is reacted. Note that the stoichiometry of magnetite affects its stability (e.g., Gorski et al., 2010), and increasing Fe(II) concentrations in magnetite theoretically increases its solubility except under extremely reducing conditions.

Magnetite has been observed to form as a product of the reaction of Fe(III) (hydr)oxides with Fe^{+2} , either when Fe(II) is directly introduced in solution (e.g., Jang et al., 2008; Yang et al., 2010) or produced by the bioreduction of Fe(III) (hydr)oxides (e.g., Hansel et al., 2003; Behrends and van Cappellen, 2005). In the simulations discussed above, magnetite is predicted to form only when ferrihydrite is the reactant. It can be shown that the reaction of Fe(II) with ferrihydrite is thermodynamically favorable at micromolar Fe^{+2} concentrations at pH ~ 5 , and much lower Fe^{+2} concentrations at pH > 5 , whereas the reaction with either hematite or goethite requires much higher Fe^{+2} concentrations. The conversion of ferrihydrite to magnetite with concurrent oxidation of UO_2 can be written as:



or, if enough UO_2 is oxidized to reach schoepite solubility:



In the latter case, the (overall) reaction is no longer dependent on the solution composition, with a $\log(K)$ value implying oxidation would readily proceed until complete $\text{UO}_{2(\text{bio})}$ and/or ferrihydrite consumption. However, magnetite precipitation is likely to be slow and/or proceed through the formation of intermediary products such as goethite and lepidocrocite (Hansel et al., 2003, 2005; Yang et al., 2010), thus limiting the extent of reaction (6) or (7). For this reason, and to keep the complexity of the modeled system manageable, the formation of magnetite is not considered further in this study.

For systems unbuffered by secondary minerals, our calculations show that oxidation is sensitive to dissolved bicarbonate and Fe concentrations, as well as speciation (e.g., reactions (1)–(3)), consistent with the thermodynamic evaluations of Ginder-Vogel et al. (2006). In our case, dissolved carbonate drives UO_2 oxidation by complexing not only strongly with U(VI) (Dong and Brooks, 2006) but also Fe(II) (Millero et al., 1995; Preis and Gamsjäger, 2002) at alkaline pH. Another potentially critical factor affecting the degree of oxidation, as discussed earlier, is the particle size of $\text{UO}_{2(\text{bio})}$. To illustrate this effect, calculations analogous to those presented above were repeated for bulk $\text{UO}_{2(\text{c})}$ as well as for UO_2 particles of various diameters (3, 5 and 10 nm). Differences in UO_2 solubility with particle size were based on Eq. (B.1) and a surface energy of 0.73 J/m² neglecting surface stress effects (Appendix B). Results show a wide range of computed U concentrations (essentially all U(VI); Fig. 2), with largest differences at the lower end of the particle-size range, as would be expected from Eq. (B.1). At pH values up to circumneutral, the precipitation of siderite and schoepite for cases of UO_2 particle sizes $< \sim 5$ nm yields lower U concentrations that mask the effect of increased solubility with decreasing particle size (Fig. 2). Recent experiments by Ulrich et al. (2008, 2009) for Fe-free systems suggest that there may not exist large differences in the solubility of bulk and biogenic UO_2 (at pH 7–8), and that aggregation may significantly limit the solubility increase predicted on theoretical grounds. However, in light of the above calculations, one cannot rule out cases when an increase in reactivity at smaller particle sizes could be overshadowed by the precipitation of secondary phases.

4. COMPETITIVE EFFECT OF SULFIDE

Because the present study targets the potential reoxidation of UO_2 produced by SRB, the effect of sulfide on UO_2 reoxidation by Fe(III) (hydr)oxides was evaluated. Sulfide can reduce U(VI) to U(IV) yielding sulfate (e.g., Ho and Miller, 1986; Mohagheghi et al., 1985) and also elemental sulfur (Beyenal et al. 2004; Hua et al., 2006). Therefore, thermodynamically, any common oxidant of sulfide and UO_2 would be expected to oxidize sulfide preferentially to UO_2 . This has been shown to be the case, for example, with

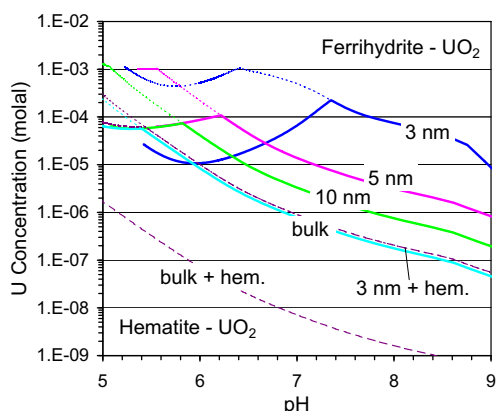
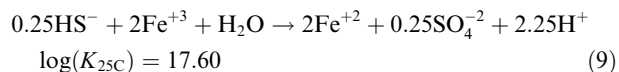
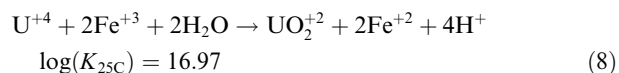


Fig. 2. Computed total dissolved U equilibrium concentrations upon oxidation of UO₂ by either ferrihydrite (solid lines) or hematite (dashed lines) as a function of pH, assuming various UO₂ particle sizes (from 3 nm particles to bulk UO₂; see text). Dotted lines show solubilities calculated assuming meta-schoepite instead of schoepite. The breaks in the curves correspond to the point when schoepite (or meta-schoepite) starts to precipitate as the pH is lowered. Note that the curve for the system with 3-nm UO₂ and hematite nearly coincides with the curve for the system with bulk UO₂ and ferrihydrite. Also note that siderite is predicted to form in the case with ferrihydrite and 3 nm UO₂ (only in this case, and over the entire pH range), significantly altering the solubility trend as function of pH once schoepite forms (see text and reaction 4). Solutions are initially fully reduced, with initial 0.1 mM Ca and 3 mM ΣCO₃ concentrations.

biogenically produced FeS (mackinawite), which appears to protect UO₂ from oxidation in systems where O₂ is the main oxidant (Abdelouas et al., 1999). Because Fe(III) (hydr)oxides are also known to oxidize sulfide (Rickard, 1974; Pyzik and Sommers, 1981; Poulton, 2003; Poulton et al. 2004), these phases are expected to preferentially oxidize sulfide over UO₂. Therefore, UO₂ reoxidation is expected to depend on the relative rate of U(IV) and sulfide oxidation by Fe(III), and also on the Fe⁺² activity in the system ($a_{\text{Fe}^{+2}}$), as discussed below.

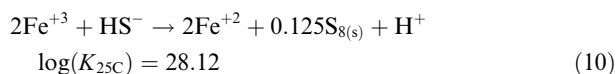
The oxidation of U(IV) and sulfide by Fe(III) can be expressed by the following two general redox reactions:



As both reactions proceed and produce Fe⁺², the least thermodynamically favored of the two, reaction (8), will reach equilibrium before reaction (9) and eventually reverse as $a_{\text{Fe}^{+2}}$ continues to increase through reaction (9). The speed at which this reversal occurs depends on the relative rate of these reactions, ranging from unimpeded reoxidation when the rate of reaction (9) is nil, to completely suppressed reoxidation once the rates of both reactions become equal. It is also evident from these reactions that any lowering of $a_{\text{Fe}^{+2}}$ would promote reoxidation. The precipitation of FeS and/or formation of strong Fe(II) sulfide

complexes (Rickard, 1995, 2006; Rickard and Luther, 2007) decreases $a_{\text{Fe}^{+2}}$, which then at least initially counters the otherwise protective role of sulfide against reoxidation.

Native (elemental) sulfur is also a common product of the Fe(III) (hydr)oxide reductive dissolution by sulfide (Poulton et al., 2004), and Sani et al. (2004) further suggested it formed in their experiments. In this case, sulfur precipitates primarily from the oxidation of sulfide by Fe(III), which can be expressed by the following general reaction:



This reaction competes for sulfide with the precipitation of FeS and also with reaction (9), and leaves more Fe(II) in solution than in the case of FeS precipitation alone, thus resulting in higher $a_{\text{Fe}^{+2}}$ and more protection against UO₂ oxidation. These reactions are obviously tightly coupled, and a higher $a_{\text{Fe}^{+2}}$ in turn favors FeS precipitation.

This somewhat complicated balance of competing reactive processes is illustrated by simple models of UO_{2(bio)} oxidation under kinetic constraints, with and without precipitation of FeS and/or sulfur (Fig. 3). The simulations are run at pH 7, using (separately) hematite and ferrihydrite, and starting with a solution of the same initial composition as considered previously (3 mM bicarbonate and 0.1 mM Ca), plus an added 1 mM sulfide (in the range of concentrations at bioremediation sites with induced SRB activity). The rate of reaction (8) is set to a constant arbitrary value ($R_8 = 10^{-10}$ mol/s). Simulations are then run varying the values for the rate of reaction (9), ranging from $R_9 = 0$ to $R_9 = 0.8 \times R_8$ (a point at which very little reoxidation is predicted). The first set of simulations is run without allowing the precipitation of FeS or sulfur (Fig. 3A, C and E). Other cases are run with precipitation of FeS with and without precipitation of sulfur (Fig. 3B, D and F). In each case, except for sulfur, the specified rates of mineral dissolution and precipitation are set to values higher than for reactions (8) and (9) (by over four orders of magnitude) such that the reaction with solids is limited only by the rates imposed on reactions (8) and (9). In the case where sulfur is allowed to form, the rate of reaction (10) is arbitrarily set to $R_{10} = 10^{-10}$ mol/s (the same as R_8), thus at a value competing with reaction (9).

When ferrihydrite is reacted, enough U is oxidized to reach the solubility of schoepite (Fig. 3E and F), and the precipitation of this phase significantly limits predicted U concentrations. When the precipitation of schoepite is suppressed in favor of meta-schoepite, predicted U concentrations are significantly higher but do not reach meta-schoepite solubility (Fig. 3C and D). It should be noted that when U concentrations are limited by schoepite precipitation (Fig. 3E and F), about three times more siderite forms than when schoepite does not precipitate (because the limiting effect of U(VI) on reactions (1)–(3) is diminished), which more strongly depletes bicarbonate in solution and further depresses U concentrations (Fig. 3E and F).

It is evident from these simple models that increasing the rate of sulfide oxidation (R_9) progressively reverses the initial UO_{2(bio)} oxidation. In the case of hematite reaction

without FeS precipitation, computed U concentrations at $R_9 = 0$ (Fig. 3A) are about six times larger than in the system without sulfide (Fig. 1A at pH 7) because of the formation of $\text{FeS}_{(\text{aq})}$, which lowers $a_{\text{Fe}^{+2}}$, driving reaction (8) further to the right. When FeS is allowed to precipitate, the decrease in $a_{\text{Fe}^{+2}}$ is even larger, yielding U concentrations at $R_9 = 0$ (Fig. 3B) about two orders of magnitude larger than in the no-sulfide system (Fig. 1A at pH 7). This effect is much less noticeable when ferrihydrite is reacted (Fig. 3C–F) because in this case, $a_{\text{Fe}^{+2}}$ is controlled primarily by siderite precipitation. As expected, the precipitation of sulfur further impedes UO_2 oxidation (Fig. 3B, D and F at $R_9 = 0.8 \times R_8$) by allowing more Fe(II) to remain in solution than in the case of FeS precipitation alone.

The intent of these numerical experiments is to highlight the delicate balance between Fe(II) and sulfide production/

consumption reactions expected to take place in real systems (through various biotic and abiotic processes). Obviously, cases at $R_9 = 0$ (no sulfide oxidation) represent an extreme case of disequilibrium that may not be relevant to most natural systems. These cases are used to illustrate the potentially profound effect of the relative rates of these reactions on $\text{UO}_{2(\text{bio})}$ reoxidation. Also note that the rates of reactions (8) and (9) encompass both electron-transfer processes and the dissolution/precipitation rates of solid phases. The distinction between the two is not being considered here, with limiting bulk rates imposed only on reactions (8) and (9) and not on mineral precipitation/dissolution. Reported dissolution rates of UO_2 vary significantly, with values measured by Ulrich et al. (2009) for biogenic and bulk UO_2 (pH ~ 8.5) in the range of $\sim 10^{-10}$ to $\sim 10^{-11}$ mol/m²/s in anoxic conditions, and $\sim 10^{-9}$ to

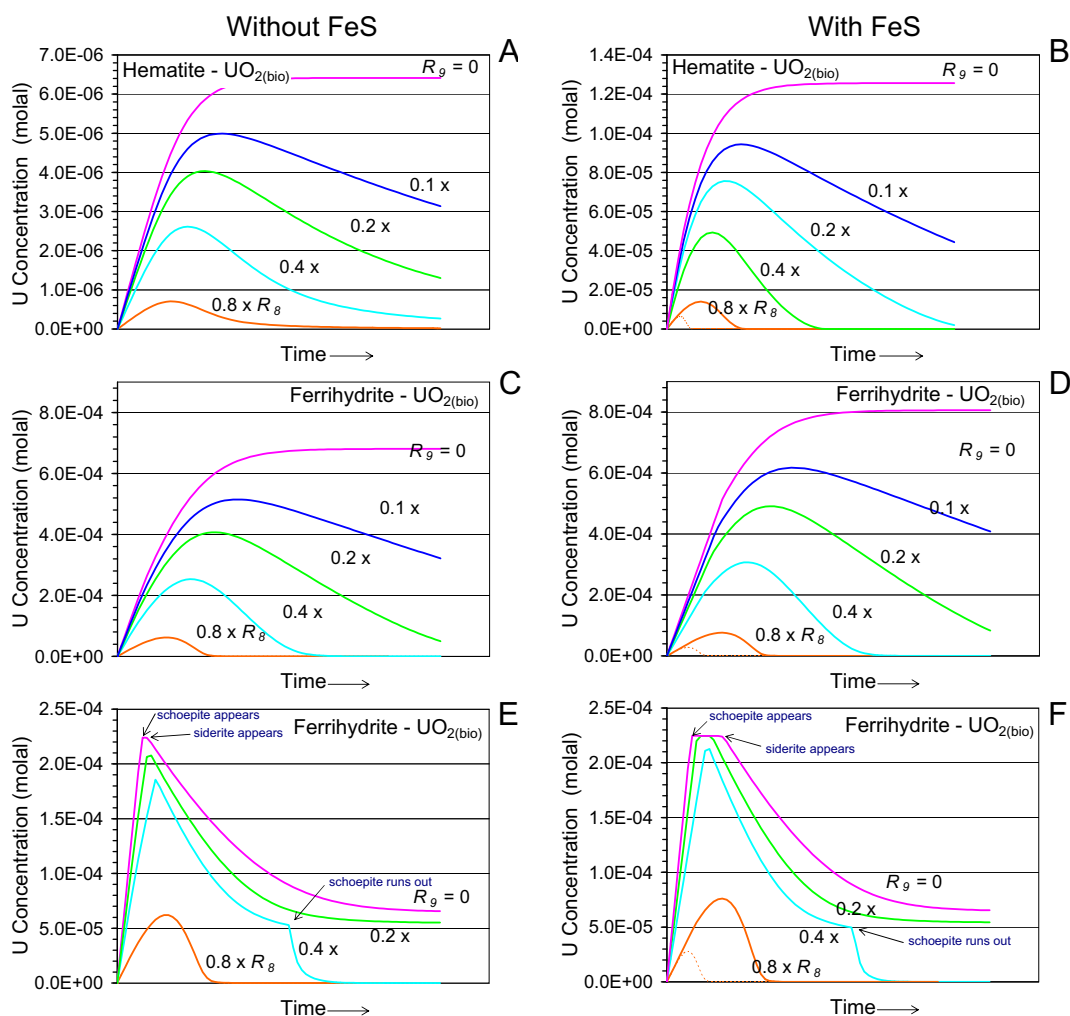


Fig. 3. Computed U solubility upon kinetic oxidation of $\text{UO}_{2(\text{bio})}$ with hematite (A and B) and ferrihydrite (C–F) at various rates of sulfide oxidation (reaction 9) relative to U(IV) oxidation (reaction 8) (see text). FeS is either suppressed (left plots) or allowed to form (right plots). Schoepite solubility is reached in B–F but this mineral is not allowed to form in B and C (in this case, meta-schoepite is allowed to form but its solubility is never reached). Sulfur is allowed to form in addition to FeS for cases with $R_9 = 0.8 \times R_8$ (dotted lines; B, D, and F), which further depresses U concentrations by allowing more Fe(II) to remain in solution (i.e., less FeS precipitates as some of the sulfide is consumed to form sulfur). The time scale is arbitrary and depends on the value of R_8 ; with $R_8 = 10^{-10}$ mol/s, equilibrium solubility is reached after up to a few days for the hematite cases and about a year for the ferrihydrite cases (i.e., at the same rate, more time is required to reach the higher solubility in the presence of ferrihydrite).

$\sim 10^{-12}$ mol/m²/s in oxidizing conditions. For hematite, using the bicarbonate dependent rate law presented by Bruno et al. (1992) and assuming 3 mM bicarbonate, the dissolution rate is calculated at $\sim 10^{-11}$ mol/m²/s. Greater bulk reductive dissolution rates for Fe(III) hydroxides were reported by Larsen and Postma (2001) (ascorbic acid solutions, pH 3), ranging from $\sim 4 \times 10^{-8}$ mol/m²/s for ferrihydrite to 4×10^{-10} mol/m²/s for goethite, respectively. By comparison, the reductive dissolution rates of these minerals in 1 mM sulfide solution, calculated from Poulton et al. (2004), range from $\sim 3 \times 10^{-9}$ to 3×10^{-11} mol/m²/s (at pH ~ 7 , with rates decreasing by about one order of magnitude for each 2 pH units increase above 7). This wide and overlapping range of rates, as well as uncertainties in reactive surface areas in natural systems, are likely to make the prediction of UO_{2(bio)} reoxidation behavior at any particular site very challenging without careful model calibration to site-specific field and laboratory data.

5. APPLICATION TO MODELING BIOREDUCTION AND CONCOMITANT REOXIDATION

Sani et al. (2004) reported on batch experiments in which biogenic UO₂ was precipitated from initially oxidized U(VI)-bearing solutions using SRB. These experiments were conducted in the presence of either hematite, goethite, or ferrihydrite, using lactate as an electron donor. Under lactate-limited conditions, it was observed that once all the lactate was consumed, UO_{2(bio)} reoxidized even though conditions remained reducing and sulfide was still present. The authors also reported that more UO_{2(bio)} reoxidized with hematite than with ferrihydrite, the reverse of the thermodynamically expected behavior.

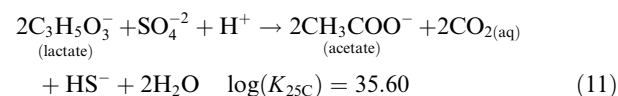
These experiments are used here to conceptualize a simulation of U(VI) bioreduction followed by reoxidation of UO_{2(bio)} in the presence of hematite or ferrihydrite. The composition of the initial solution is based on experimental conditions (Sani et al., 2004), including 90 μ M U(VI), 20 mM SO₄²⁻, 30 mM lactate, 30 mM PIPES at pH 7.2, 0.4 mM Ca, and 3 mg/L (initial) total cell protein. The dissolved carbonate concentration in the simulations is set to initially reflect equilibrium with atmospheric CO₂ (~ 0.17 mM), then allowed to evolve as a function of lactate degradation. Initial amounts of hematite (7.25 mM) and ferrihydrite (1.8 mM) are set to the actual amounts used in the experiments. These amounts were determined by Sani et al. (2004) to yield the same available surface area in both the hematite and ferrihydrite experiments.

Here, the aim is to model U bioreduction followed by reoxidation, thus to simulate a reaction network that can express concomitant U bioreduction and reoxidation in the presence of Fe(III) (hydr)oxides. The reduction phase considers microbial sulfate reduction (e.g., Lovley and Phillips, 1992) together with the reductive dissolution of Fe(III) solids, which is assumed to be initially microbially mediated (e.g., Lovley, 1991) as well as abiotically driven by sulfide. More sophisticated quantification of microbial Fe(III) reduction dynamics factors, such as effects of microbe-mineral contact extent and mineral surface area (e.g., Bonneville et al. 2006; Roden, 2003), are not considered here in order to

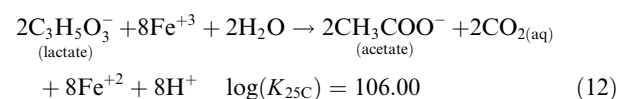
minimize the risk of model over-parameterization, and because these are deemed second-order effects in the present case (where abiotic Fe reduction dominates). Note that U bioreduction models have been presented by others, some including the microbial reduction of Fe(III), sulfate and U(VI) (e.g., Li et al., 2009; Fang et al., 2009), some including nitrate, sulfate, and U(VI) bioreduction (e.g. Luo et al., 2007), while others looking at U(VI) biotic reduction coupled to organic matter oxidation (e.g., Stewart et al., 2011). None of these models, however, consider the concomitant reoxidation of UO_{2(bio)} by Fe(III) hydroxides.

In their paper, Sani et al. (2004) presented a set of postulated reactions including, in addition to the reoxidation of UO₂ by Fe(III) minerals, the reduction of sulfate by microbial degradation of lactate to acetate and the reduction of Fe(III) and U(VI) by sulfide with sulfur as one of the end products. These authors inferred reaction stoichiometries from the molar lactate to sulfate utilization ratios and lactate consumption to acetate production ratios observed in their experiments. They also inferred reductive dissolution of Fe(III) hydroxides by sulfide from limited sulfide concentration data collected during the experiments (although sulfur was not directly measured) and suspected that the bioreduction of U(VI) and sulfate would be mostly preferential to the bioreduction of solid-phase Fe(III). On the basis of their observations, the following main aqueous reactions network is postulated to capture the U(VI) reduction and reoxidation behavior in their experiments:

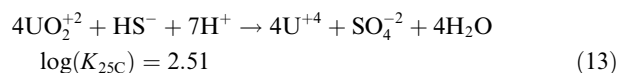
Sulfate reduction (biotic)



Fe(III) reduction (biotic)



U(VI) reduction (combined abiotic and biotic, see text)



U(IV) oxidation by Fe(III) (abiotic)

Reaction (8)

HS⁻ oxidation by Fe(III) (abiotic)

Reaction (9)

These reactions are coupled to the dissolution and/or precipitation of solid phases including UO_{2(bio)}, hematite or ferrihydrite, siderite, FeS, and sulfur, and the model is run as a closed system consistent with the experimental setup. The dissolution and precipitation reactions of solid phases are assumed to be limited only by the rate of the above aqueous reactions, except for sulfur, for which the precipitation needed additional kinetic constraints to yield results consistent with the experiments.

The sulfate and initial iron reduction reactions (11) and (12) are biotic, with reaction rates modeled using a conventional dual-Monod rate law with biomass growth:

$$R = qC_b \frac{C_D}{k_D + C_D} \frac{C_A}{k_A + C_A} f_G \quad (14)$$

with

$$R_b = Y_b R - bC_b \quad (14a)$$

$$f_G = (1 - Q/K) \quad (14b)$$

Here, subscripts *A*, *D*, and *b* stand for electron acceptor, electron donor, and biomass, respectively, *C* is concentration, *k* is half saturation constant in units of *C*, *q* is the rate of maximum substrate utilization (in units of moles per time, per biomass), *Y* the microbial yield coefficient (in units of biomass per substrate), and *b* the cell decay rate (in units of per time). Values of *q*, *Y_b*, *k_D* and *k_A* were estimated by calibration to the experimental data (Table 1). The affinity term, *f_G*, incorporates the reaction ion activity product, *Q*, and equilibrium constant, *K*. This term varies between 1 (away from equilibrium) to 0 when the reaction reaches thermodynamic equilibrium.

The rates of reactions (8)–(10) are expressed as

$$R = rf_G \quad (15)$$

where *r* (in units of moles per time) is assumed to be constant. These reactions operate close to equilibrium and can reverse (using the same value of *r*) if the reaction *Q* becomes greater than *K* (Eq. (14b)). Note that in the case of reversal, for simplicity we keep the value of *f_G* between 0 and 1 as in the forward direction (thus *f_G* becomes $(1 - K/Q)$ and the sign of *R* is reversed).

The U(VI) reduction (reaction 13) was initially tested using Eq. (15) and a fast (nonlimiting) rate, thus allowing this reaction to proceed at a rate limited only by the rate of sulfide generation (i.e., by the rate of reaction 11). However, when doing so, modeled U(VI) concentrations were found to decrease at a much faster rate, relative to lactate depletion, than observed by Sani et al. (2004). Two alternative approaches were tested to capture the experimental data. The first and simplest approach, adopted here, was to limit the rate of reaction (13) by lactate, by simply multiplying the rate of this reaction by a Monod function in terms of the lactate concentration (*C_D*),

$$R = r \frac{C_D}{k_D + C_D} f_G \quad (16)$$

and adjusting *r* and *k_D* in this expression to reproduce observed trends. This was done without adding another reaction for U(VI) reduction by lactate, and allowed the

forward rate of reaction (13) to be lactate-limited and smaller than that of reaction (11), as suggested by the experimental results. The reaction is allowed to operate under thermodynamic constraints (of reaction (13)) in the reverse direction (using the same value of *r* but without the Monod term), although in the present simulations this reaction was not predicted to reverse. Because much more lactate is oxidized by sulfate than by U(VI), simply limiting reaction (13) on lactate simplifies the reaction network while having a negligible effect on the total lactate concentration. The second approach tested was to include in the reaction network, in addition to reaction (13), one more U bioreduction reaction with lactate and U(VI) as the electron donor and acceptor, respectively, with a rate given by Eq. (14). However, doing so required fitting additional Monod kinetic parameters, without significantly improving the match to experimental data. Therefore, the first approach minimizing the model parameterization was preferred.

In contrast, the oxidation of U(IV) and HS[−] by Fe(III) (reactions 8–10) was considered to be fully abiotic and limited only by thermodynamic constraints (Eq. (15)). The model was run for a simulated period (~60 days) covering the length of the experiments, and kinetic parameters were adjusted (Table 1) to capture observed trends and magnitudes of lactate, acetate, sulfate, HS[−], and U(VI) concentrations (Fig. 4).

It should be noted that the experiments of Sani et al. (2004) show noticeable time lags before significant lactate degradation and U(VI) reduction occur, especially in the case of experiments with hematite. This is likely the result of U(VI) toxicity to SRB (e.g., Sani et al., 2006). Microbial growth kinetic models to simulate lag time in the presence of inhibitors have been proposed (e.g., Sengor et al., 2009; Ginn, 1999). However, these models require validation using multiple increasing concentrations of the inhibitor for both pure and mixed culture data, which was not within the scope of Sani et al. (2004). Therefore, mechanistic modeling of these time lags could not be considered in the present study. Instead, the experimental results of Sani et al. (2004) on Fig. 4 were shifted in time by about 26 days in the hematite case, and 3 days in the ferrihydrite case, to reflect concentration profiles starting at the point when significant bioreduction began.

Another process not incorporated in the simulations shown on Fig. 4 is the adsorption of Fe⁺² and U(VI) onto Fe(III) (hydr)oxides (e.g., Appelo et al., 2002; Jang et al., 2007b and references therein). Test simulations incorporating

Table 1

Calibrated kinetic parameters used for the simulation of U(VI) bioreduction in the presence of (separately) hematite and ferrihydrite (Fig. 4; see text).

Process	Reaction	Rate law	<i>q</i> (mol/s/mg _{cells}) or <i>r</i> (mol/s)	<i>k_D</i> (mol/L)	<i>k_A</i> (mol/L)	<i>Y_b</i> (mg _{cells} /mol)	<i>b</i> (1/s)
Sulfate bioreduction	(11)	(14)	10 ^{−8}	2 × 10 ^{−2}	2 × 10 ^{−2}	1600	10 ^{−8}
Fe(III) bioreduction	(12)	(14)	10 ^{−11}	2 × 10 ^{−2}	10 ^{−20} ^a	1600	10 ^{−8}
U(VI) bioreduction	(13)	(16)	8 × 10 ^{−11}	4 × 10 ^{−2}			
Fe(III) reduction by HS	(9)	(15)	1.8 × 10 ^{−11}				
U(IV) oxidation by Fe(III)	(8)	(15)	2 × 10 ^{−11}				
Sulfur precipitation/dissol. ^b	(10)	(15)	1.4 × 10 ^{−11}				

^a Rate assumed essentially unlimited by the electron-acceptor (donor-limited experiments).

^b Rate incorporates (and assumes) a constant surface area.

surface complexation using data from these sources yielded essentially the same results as shown in Fig. 4, except with a calibrated rate for U(VI) reduction (reaction 13) about ~70% larger than in the model without sorption. The solutions in the experiments of Sani et al. (2004) were equilibrated with surfaces prior to inoculation, and all experiments were conducted using Fe(III) hydroxide amounts that reflected the same total available surface area. For these reasons, for the sake of simplicity, and because the difference in calibrated rates was considered small (compared to order-of-magnitude spans typical of reaction rates, as shown for the published dissolution rates of UO_{2(bio)} and Fe(III) minerals discussed earlier), sorption was not considered further. This also avoided potential pitfalls and inconsistencies between various available sorption constant datasets.

Simulation results are shown in Fig. 4. Even though the experimental profiles of sulfate, lactate, and acetate (as well as biomass growth, not shown) provided a good constraint on overall degradation rates, it is likely that the set of input parameter values for these simulations is non-unique given the number of calibrated parameters (Table 1). It should also be noted that these parameters pertain to the simulated experiment and should not be inferred to be applicable under different conditions or at field scales.

As in the numerical experiments presented earlier, model results show the oxidation of HS⁻ by Fe(III) (reactions 9 and 10) directly competing with UO₂ reoxidation (reaction 8) as Fe(III) oxidizes HS⁻ preferentially to UO_{2(bio)}. Sulfide profiles were not measured by Sani et al. (2004), however, these authors reported final sulfide concentrations of the same order of magnitude as simulated here (0.92 mM vs. maximum predicted ~0.5 mM in the hematite case; 6.92 mM vs. maximum predicted ~12 mM in the ferrihydrite case). Measured Fe(II) (aqueous) concentrations by Sani et al. (2004) were below detection limits (<0.11 μM), which may have resulted from oxidation prior to analysis. In the simulations, both FeS and S are predicted to precipitate, with a molar FeS/S ratio about 2 to 1 in both the hematite and ferrihydrite cases. Siderite does not form, even in the ferrihydrite case, because the amount of ferrihydrite available is too small to yield enough iron in solution to trigger siderite precipitation.

Except at very early times in the experiment with hematite, most of the Fe reduction predicted by the model is abiotic (reactions 9 and 10) (Fig. 4B and E). Because the initial amount of ferrihydrite used in the experiment is smaller (1.8 mM) than that of hematite (7.25 mM) (to represent equal total surface area in each case), ferrihydrite is rapidly

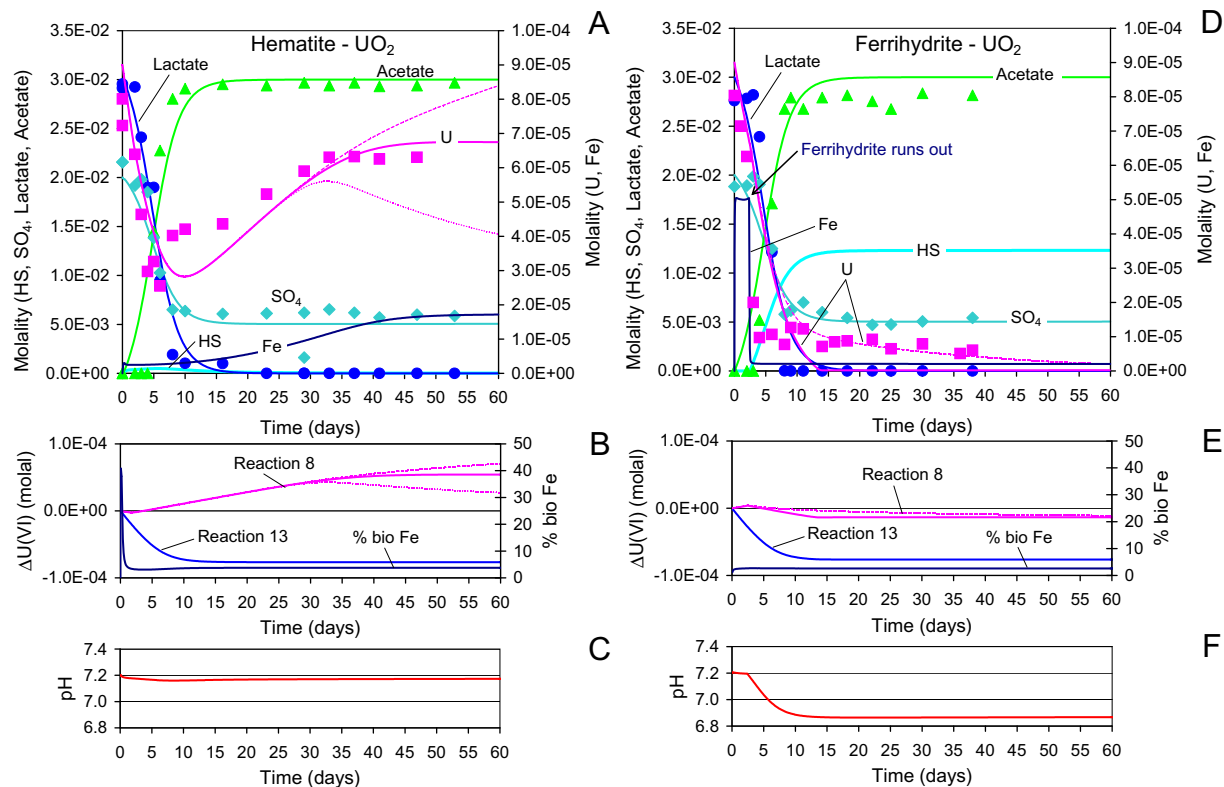


Fig. 4. Bioreduction of U(VI) in the presence of either hematite (A–C) or ferrihydrite (D–F). Symbols are experimental data points from Sani et al. (2004), and solid lines are model simulations conducted using the parameter values listed in Table 1. Dashed and dotted lines are computed U concentrations for various sensitivity cases of Fe(III) reactivity towards sulfide: hematite case, dashed line for $R_9 = 0.5 \times$ original, dotted line for $R_9 = 2 \times$ original; ferrihydrite case, dashed line for R_9 and $R_{10} = 0.02 \times$ original. Middle plots (B and E) show the change in U(VI) concentrations computed from reactions 8 and 13 (relative to the initial concentration of 90 mM), and the computed percentage of microbially produced Fe(II) (from reaction 12, relative to Fe(II) from all reactions). The greater reactivity of sulfide with ferrihydrite results in full dissolution (exhaustion) of this mineral (D), leading to apparently less U(IV) reoxidation than with hematite (reverse of the expected thermodynamic behavior). Computed total dissolved carbonate concentrations follow (overlap) the computed trends for acetate.

fully consumed (Fig. 4D), primarily by reaction with sulfide (Fig. 4E). Before this point, reoxidation starts to kick in (reaction 8), however this reaction reverses soon after ferrihydrite runs out (Fig. 4E, after ~4 days). As reaction (8) reverses, Fe(II) starts to reduce U(VI) (with Fe(III) recycled by the other Fe(III)-consuming reactions) until some steady state is reached (Fig. 4E). This contrasts with the hematite case, in which the reoxidation is initially not favorable (Fig. 4B, before ~3 days) but later becomes favorable as hematite remains available. The exhaustion of ferrihydrite is followed by a slight pH drop to ~6.9 from the initial value buffered around 7.2 (Fig. 4F), which is not predicted in the hematite case (Fig. 4C).

Computed dissolved carbonate concentrations from the degradation of lactate are elevated, and follow (overlap) the computed acetate trends (Fig. 4) with final values around 30 mM, dictated by the 1:1 acetate to inorganic carbon ratios in reactions (11) and (12). The increase in dissolved carbonate concentrations (yielding elevated $\log(p_{\text{CO}_2})$ values around -1 in this closed system) causes more U(VI)–CO₃ complexing, which, in the case of reoxidation by hematite (Fig. 4A), yields close to two order-of-magnitude higher U(VI) concentrations than computed earlier (Fig. 1A) for lower carbonate concentrations (3 mM) more typical of natural groundwaters. In the ferrihydrite case (Fig. 4D), the carbonate effect is less noticeable because ferrihydrite is a stronger oxidant (than hematite) and once this mineral runs out, the added effect of oxidation by Fe(III) vanishes. These results support concerns raised by others (Tokunaga et al., 2008) about U bioremediation potentially increasing the risk of UO_{2(bio)} reoxidation by the production of dissolved carbonate from the degradation of the organic electron donor.

6. CONCLUSIONS

The numerical experiments and biogeochemical simulations presented here capture important thermodynamic and (bulk) kinetic constraints affecting the reoxidation of biogenic UO₂ by Fe(III) (hydr)oxides under sulfate-reducing conditions. Based on these constraints alone, this study highlights the quite complicated and delicate reactive balances taking place in the U–Fe–S–Ca–CO₃ system. The reoxidation of biogenic UO₂ is subject to competing rates of Fe(II) and sulfide production and/or consumption by various biotic and abiotic reactions. In particular, the relative rate of U(IV) versus sulfide oxidation by Fe(III) appears to be an important factor affecting UO₂ reoxidation. The system is also very sensitive to the activity of Fe⁺², which itself depends on rates of Fe(III) reduction, and rates and types of reacting Fe(II) minerals. These various and complex reaction feedbacks are compounded by large uncertainties on the solubility of biogenic UO₂, and are expected to make field-scale predictive modeling work regarding reoxidation very difficult without careful model calibration to site-specific field and laboratory data. In addition, such a chemically complex system would need to be placed quantitatively in the context of natural diffusive, dispersive, and advective transport *in situ*, with mixing

limitations associated with real motion further complicating the model predictive capability.

The models developed in this study are simplified, compared to natural systems, and notably do not incorporate actual mechanistic reoxidation reaction mechanisms, many of which are still poorly understood. Added levels of complexity, such as for example surface-catalyzed reactions (e.g., Ginder-Vogel et al., 2010), consideration of molecular U(IV) (Fletcher et al., 2010), and substitution of U(VI)/U(IV) in Fe(II)/Fe(III) phases (Nico et al., 2009; Stewart et al., 2009; O'Loughlin et al., 2010), are likely to add to the difficulty of developing models that can accurately predict both the timing and magnitude of UO₂ reoxidation for the design of actual bioremediation systems. Nevertheless, simpler thermodynamic and kinetic models as developed in this study allow the evaluation of general reoxidation behavior in a qualitative or semi-quantitative manner, which increases the understanding of this complex system towards planning effective remedial activities at contaminated sites.

ACKNOWLEDGMENTS

Funding for this research was provided by the U.S. Department of Energy, Office of Science, Subsurface Biogeochemical Research (SBR) Contract DE-FG02-07ER-64366. This work was also partially supported by the US Department of Energy and LBNL under Contract No. DE-AC02-05CH11231. We thank two anonymous reviewers and particularly P. Van Cappellen and his group for their valued constructive review comments. Editorial support by D. Hawkes is also greatly appreciated.

APPENDIX A

The $\log(K)$ values shown in Table A1 are for dissociation and dissolution reactions in terms of the following aqueous species: U⁺⁴, UO₂⁺², Fe⁺², Fe⁺³, HS[−], SO₄^{−2}, H⁺, Ca⁺², CO₃^{−2}, and H₂O, unless indicated otherwise. For species other than U, data are given only for selected potentially dominant species, and for other species of interest to convert reactions.

Table A1

<i>U(VI) species</i>	$\log(K_{25^\circ\text{C}, 1\text{bar}})$	Ref.
(UO ₂) ₁₁ (CO ₃) ₆ (OH) ₁₂ ^{−2}	−36.12	1
(UO ₂) ₂ (OH) ₂ ⁺²	5.657	1
(UO ₂) ₂ CO ₃ (OH) ₃ [−]	0.916	1
(UO ₂) ₂ OH ⁺³	2.729	1
(UO ₂) ₃ (CO ₃) ₆ ^{−6}	−53.88	1
(UO ₂) ₃ (OH) ₄ ⁺²	11.96	1
(UO ₂) ₃ (OH) ₅ ⁺	15.62	1
(UO ₂) ₃ (OH) ₇ [−]	32.20	2
(UO ₂) ₃ O(OH) ₂ (HCO ₃) ⁺	−0.583	1
(UO ₂) ₄ (OH) ₇ ⁺	21.99	1
UO ₂ OH ⁺	5.217	1
UO ₂ (OH) ₂ ⁰	12.15	2
UO ₂ (OH) ₃ [−]	20.25	2
UO ₂ (OH) ₄ ^{−2}	32.40	2

UO ₂ CO ₃ ⁰	-9.94	2
UO ₂ (CO ₃) ₂ ⁻²	-16.61	2
UO ₂ (CO ₃) ₃ ⁻⁴	-21.84	2
CaUO ₂ (CO ₃) ₃ ⁻²	-27.18	3
Ca ₂ UO ₂ (CO ₃) ₃ ⁰	-30.70	3
<i>U^(IV) species</i>		
U(CO ₃) ₄ ⁻⁴	-35.05	1
U(CO ₃) ₅ ⁻⁶	-33.82	1
UOH ⁺³	0.541	1
U(OH) ₂ ⁺²	1.09	2
U(OH) ₃ ⁺	4.69	2
U(OH) ₄ ⁰	9.98	2
<i>U^(VI) and U^(IV) solids</i>		
Schoepite (UO ₃ ·2H ₂ O)	4.844	1
Meta-schoepite	5.6	9
UO _{2(am)}	1.48	2
UO _{2(3nm)} (UO _{2(bio)})	1.42	4
UO _{2(5nm)}	-1.10	4
UO _{2(10nm)}	-2.99	4
Uraninite (UO _{2(c)})	-4.88	2
<i>Fe^(III) and Fe^(II) species</i>		
FeOH ⁺	9.315	1
Fe(OH) ₂ ⁰ (or FeO ⁰)	20.405	1
FeCO ₃ ⁰	-5.450	1
FeHCO ₃ ⁺	-11.799	1
FeS ⁰	2.2	7
FeSO ₄ ⁰	-2.200	1
Fe(OH) ₂ ⁺ (or FeO ⁺)	5.650	1
Fe(OH) ₃ ⁰ (or HFeO ₂ ⁰)	12.018	1
Fe(OH) ₄ ⁻ (or FeO ₂ ⁻)	21.620	1
<i>Fe^(III) and Fe^(II) solids</i>		
Hematite (Fe ₂ O ₃)	0.109	1,1a
Goethite (FeOOH)	0.176	5
Ferrihydrite (Fe(OH) ₃)	3.116	5
Magnetite (FeO·Fe ₂ O ₃)	10.472	1,1a
Siderite (FeCO ₃)	-10.521	1,1a,6
Mackinawite (FeS _(mc))	-3.5	7
<i>Other relevant reactions</i>		
Calcite (CaCO ₃)	-8.480	1
CO _{2(g)}	-18.142	1
CO _{2(aq)} (or H ₂ CO ₃ ⁰)	-16.674	1
HCO ₃ ⁻	-10.329	1
<i>Redox reactions</i>		
U ⁺⁴ + 2H ₂ O ↔ UO ₂ ⁺² + 4H ⁺ + 2e ⁻	-9.049	1
UO ₂ ⁺ ↔ UO ₂ ⁺² + e ⁻	-1.473	1
Fe ⁺² ↔ Fe ⁺³ + e ⁻	-13.011	1
HS ⁻ + 4H ₂ O ↔ SO ₄ ⁻² + 9H ⁺ + 8e ⁻	-33.690	1
HS ⁻ ↔ S _(s) + H ⁺ + 2e ⁻	2.096	1
2H ₂ O ↔ O _{2(aq)} + 4H ⁺ + 4e ⁻	-86.003	1
2H ₂ O ↔ O _{2(g)} + 4H ⁺ + 4e ⁻	-83.105	1
C ₃ H ₅ O ₃ ⁻ (lactate) + 6H ₂ O ↔ 3CO ₃ ⁻² + 17H ⁺ + 12e ⁻	-61.636	8
C ₂ H ₃ O ₂ ⁻ (acetate) + 4H ₂ O ↔ 2CO ₃ ⁻² + 11H ⁺ + 8e ⁻	-45.916	8

¹SNL (2007), from Shock et al. (1997) and for U, within error margins of Guillaumont et al. (2003); ^{1a}Data for solids from Helgeson et al., 1978; ²Guillaumont et al. (2003); ³Dong and Brooks (2006); ⁴This study; ⁵Adjusted from SNL(2007) for consistency with Majzlan et al. (2004); ⁶Also consistent with Preis and Gamsjäger (2002); ⁷Rickard (2006); ⁸Calculated from Shock (1995) and Shock et al. (1997). ⁹Gorman-Lewis et al. (2008).

APPENDIX B

Particle sizes in the range of nanometers affects thermodynamic stability because of surface energy and surface stress effects (e.g., Zhang and Banfield, 1998). The surface energy can be regarded as the excess energy generated by un-terminated bonds at the surface of a solid (unbalanced surface charge). The surface stress is caused by deviations in the electronic structure of the surface (compared to the bulk solid) to compensate for the unbalanced charge. There is no consensus on a unique method to estimate the solubility dependence on particle size for nanoparticles. Here we follow the approach presented by Zhang and Banfield (1998) and Banfield and Zhang (2001). Assuming the surface stress equals $1 \times$ the surface free energy (a first approximation), spherical particles, and re-expressing standard thermodynamic expressions for surface free energy in terms of stability constant K (with $\Delta G = -RT \ln(K)$), the increase in solubility product for a nanoparticulate phase, relative to its bulk form, can be expressed as a function of particle size as:

$$\Delta \log(K_{sp}) = \frac{6M\gamma}{D\rho} \frac{1}{2.303RT} + \frac{4M\gamma}{D\rho} \frac{1}{2.303RT} \quad (\text{B.1})$$

where, γ is the surface energy (here taken as 0.73 J/m² = N/m, see below), M and ρ are the molecular weight (270.03 g/mol) and density (10.96×10^6 g/m³; Fink, 2000) of UO₂, respectively, D is the particle diameter (m), R is the gas constant (8.314 J/K/mol) and T is absolute temperature (298.15 K). This equation is intentionally written using two similar terms to distinguish between contributions from surface energy only (first term) and added stress effects (second term), yielding results as shown in Fig. B1. Note that this equation does not account for changes in the volume to surface area ratio with particle size. Doing so would yield a factor of 4 instead of 6 in the first term of Eq. (B.1) (i.e., the Ostwald-Freundlich equation; e.g., Wu and Nancollas, 1998, and references therein). Assuming different particle shapes can also yield different multiplication factors in Eq. (B.1) (e.g., Wu and Nancollas, 1998).

The surface energy value of 0.73 J/m² for UO₂ was estimated starting from a value computed by Skomurski et al. (2006) (1.194 J/m² for the (100) face, consistent with cubic UO₂) then lowered to consider hydration based on the same average ratio of anhydrous to hydrous surface energies reported by Mazeina and Navrotsky (2007) for Fe (hydr)oxides (on the basis of similar enthalpies of water adsorption; Skomurski et al., 2008). This value is in the range of mean surface energies at 25 °C ($\sim 0.85 \pm 0.6$ J/m²) obtained from correlations presented by Hall and Mortimer (1987), and yields a solubility consistent with that given by Guillaumont et al. (2003) for UO_{2(am)} at a particle size of about 3 nm when surface stress is neglected (and thus assumed to be compensated by hydration) (Fig. B1). For comparison, $\Delta \log(K_{sp})$ values calculated using the data and method presented by Opel et al. (2007) are also shown on Fig. B1. The equation applied by these authors is essentially the Ostwald-Freundlich equation (no stress term and a factor of 4 in the first term of Eq. (B.1)) with an implicit calculation

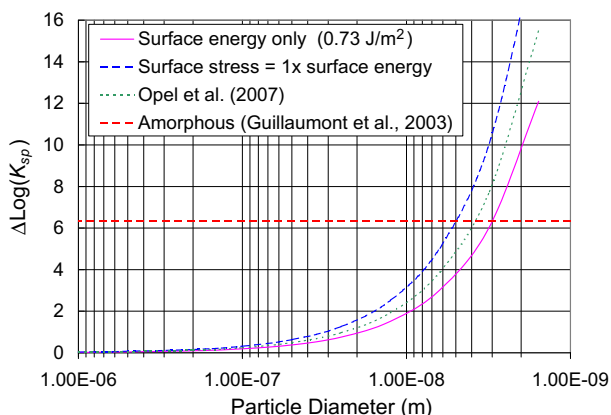


Fig. B1. Increase in UO_2 solubility product (relative to bulk $\text{UO}_{2(c)}$) with particle size, calculated using Eq. (B.1) assuming only surface energy effects (solid line) and added stress effects (dashed line) (see text). The dotted curve is calculated after Opel et al. (2007). The horizontal dashed line corresponds to the $\text{UO}_{2(\text{am})}$ solubility value reported by Guillaumont et al. (2003).

of surface energy from ionic radius data (after Schindler, 1967; yielding a surface energy value of $\sim 1.4 \text{ J/m}^2$).

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